

Logistic Regression for Classification

Theory: Used for classification tasks to predict probabilities between 0 and 1. Outputs a binary category (e.g., Yes/No) based on a threshold.

Sigmoid Function (Activation)

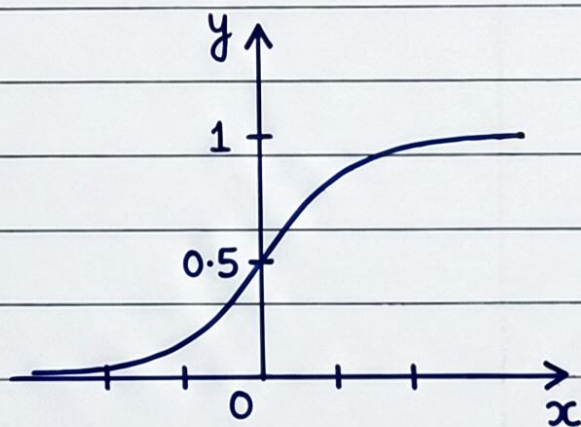
$$g(z) = \frac{1}{1 + e^{-z}}, \text{ where } e = \text{Euler's number} \approx 2.718$$

It "squashes" any real number input (z) into the range $[0, 1]$.

Hypothesis Function

$$h_{\theta}(x) = g(\theta_0 + \theta_1 x)$$

$$h_{\theta}(x) = \frac{1}{1 + e^{-(\theta_0 + \theta_1 x)}}$$

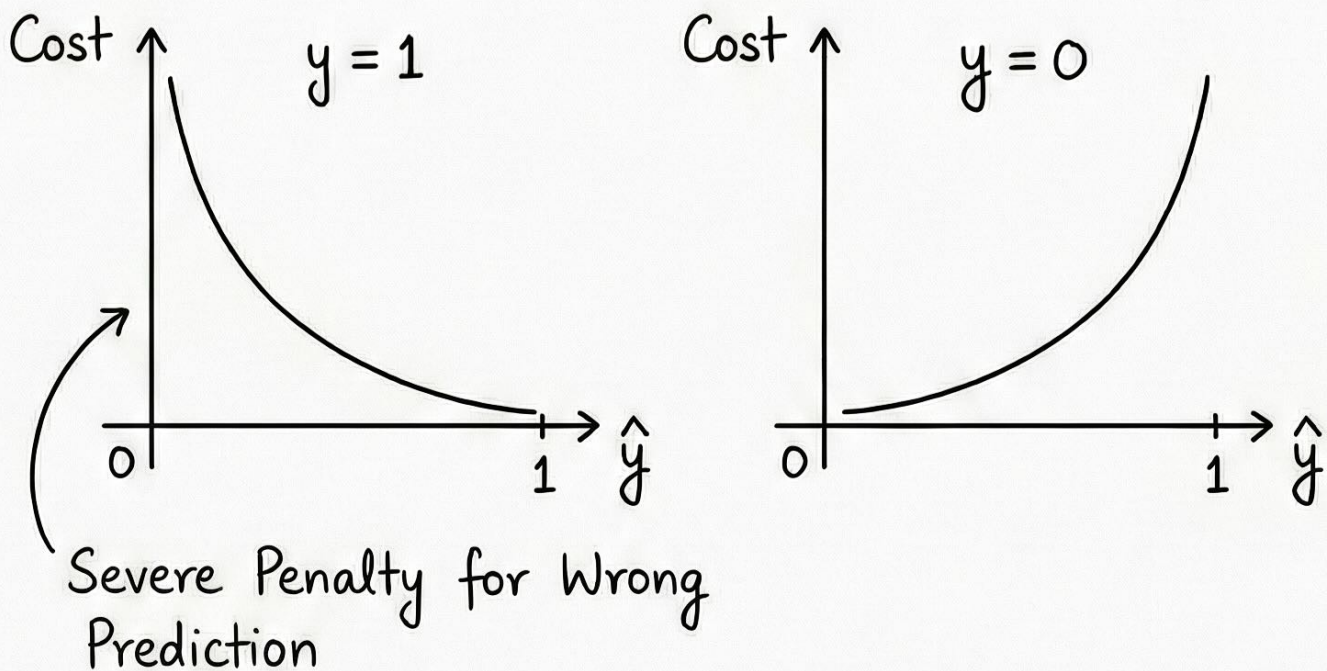


Logistic Regression – Cost Function

Actual label: $y \in \{0, 1\}$

Predicted output: $\hat{y} = h_{\theta}(x)$

$$\text{Cost}(\hat{y}, y) = \begin{cases} -\log(\hat{y}) & \text{if } y = 1 \\ -\log(1 - \hat{y}) & \text{if } y = 0 \end{cases}$$



Why not squared error? The cost function becomes non-convex, making it hard to find the global minimum. This cost function (log loss) heavily penalizes confident wrong predictions (e.g., predicting $\hat{y} \approx 0$ when $y = 1$).

Combined Cost Function (Log Loss / Binary Cross-Entropy)

$$\text{Cost}(\hat{y}, y) = -[y \cdot \log(\hat{y}) + (1-y) \cdot \log(1-\hat{y})]$$

(This one line handles both $y=1$ and $y=0$ cases)

$$J(\theta) = -\frac{1}{m} \sum_{i=1}^m [y^{(i)} \cdot \log(\hat{y}^{(i)}) + (1-y^{(i)}) \cdot \log(1-\hat{y}^{(i)})]$$

This formula averages the cost (log loss) over all 'm' training examples. It is also known as Binary Cross-Entropy.